

ERONICION PRESLEY

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EDUCATION

Northeastern University, Khoury College of Computer Science, Boston, MA

Sept 2024-Present

Master of Science, Computer Science

Relevant Courses: Programming Design Paradigm, Algorithms, Foundations in AI, ML Ops, Building Scalable Distributed Systems

Loyola-ICAM College of Engineering and Technology, Chennai, India

Jun 2018-Jun 2022

Bachelor of Technology, Information Technology

Relevant Courses: Data Structures, Design and Analysis of Algorithms, Big Data Analytics, Database Management

TECHNICAL SKILLS

Programming Proficiencies: Python, SQL, JavaScript (ES6+), TypeScript, Java, C++, HTML/CSS, Go

Frameworks & Libraries: React, Node.js, Express, Next.js, LangChain.js, OpenAI API

Databases: PostgreSQL, MySQL, MongoDB, Oracle

Cloud & DevOps: AWS, Kafka, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD Pipelines

Data & AI Tools: OpenAI API, LangChain.js, Hugging Face Transformers, TensorFlow, Apache Beam, Apache Airflow

Core Competencies: Data Validation & Quality, REST API Design, Object-Oriented Programming, Data Pipeline Engineering, SDLC

WORK EXPERIENCE

Capgemini, Software Analyst, India

Oct 2022-Jun 2024

- Built automated data pipelines processing 50K+ daily records via REST APIs with retry logic, reducing manual work by 30%
- Deployed CI/CD pipelines using Jenkins with automated testing, monitoring dashboards, and alerting, achieving 99.5% production uptime
- Architected API integration layer connecting 5+ enterprise services with rate limiting and data validation, improving throughput by 40%
- Developed internal automation scripts in JavaScript for log analysis, incident resolution, and automated reporting workflows

Pinaca Labs Pvt Limited, Machine Learning Engineer Intern, India

Dec 2021-Mar 2022

- Built end-to-end Python data pipeline for preprocessing multilingual datasets of 1M+ records with data validation and quality checks
- Optimized data processing workflows through batch processing and distributed computing, reducing inference latency by 35%
- Deployed production ML models using Docker and Flask APIs with monitoring and logging for performance tracking

PROJECTS

Crossword Solver

April 2026 - May 2026

- Built AI crossword solver combining FAISS vector retrieval over 1M+ clue-answer pairs with parallel LLM-based reasoning
- Implemented loopy belief propagation with damping and multi-pass re-querying, achieving 100% accuracy on NYT 15x15 puzzles
- Engineered parallel candidate generation pipeline with 10 concurrent threads, solving full-size crosswords in under 30 seconds
- Designed React frontend with Flask SSE backend streaming real-time solve events, visualizing agent reasoning and grid updates

Boston 311 AI System – MLOps Capstone Project

Sept 2025 - Dec 2025

- Engineered scalable data pipeline processing 500K+ service requests using Apache Beam, Apache Airflow, and DVC for version control
- Developed ML prioritization system with XGBoost and MLflow, achieving 85% accuracy through feature engineering and optimization
- Designed PostgreSQL database schema with optimized indexing for efficient querying of high-volume time-series data
- Built full-stack interface with Next.js frontend and FastAPI backend, integrating Gemini API for a RAG-powered chatbot

Distributed Job Scheduler

June 2026 - July 2026

- Built fault-tolerant distributed job scheduler in Go with Kafka pipeline ensuring zero duplicate execution via consumer groups
- Containerized with Docker Compose, authored Kubernetes manifests for 3-replica deployment and Terraform module for AWS VPC
- Engineered React frontend with real-time auto-polling served by Go HTTP server, with full job lifecycle tracking in PostgreSQL

CERTIFICATES

- **AWS Certified Cloud Practitioner:** Validated knowledge of AWS cloud concepts, core services, security, architecture fundamentals.
- **ServiceNow Certified System Administrator:** Skilled in managing instances, workflows, and configurations with CSA + Vancouver Delta certifications.